

Math 242, S09
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given a first-order differential equation

$$y' = 2e^{-y}.$$

(a) Verify that $y(x) = \ln(2x + C)$ defines a one-parameter family of solutions of the given differential equation.

Solution: $y'(x) = \frac{1}{2x+C} \cdot 2 = \frac{2}{2x+C}$

$$2e^{-y} = 2 \cdot e^{-\ln(2x+C)} = \frac{2}{2x+C}$$

$$\Rightarrow y'(x) = 2e^{-y} \quad \text{Thus } y(x) = \ln(2x+C)$$

defines a one-parameter family of solution
of the given differential equation

(b) Determine a value of the constant C so that $y(x)$ satisfies the initial condition $y(2) = 0$.

Solution: $y(2) = 0 \Rightarrow$

$$\ln(2 \cdot 2 + C) = 0$$

$$\ln(4+C) = 0$$

$$4+C = 1$$

$$C = -3$$

$$\Rightarrow y(x) = \ln(2x-3)$$